

Keywords: transformer expressivity • succinctness • temporal logic • EXPSPACE-completeness

Transformers are Inherently Succinct

Fixed-Precision Transformers Are Exponentially More Compact Than RNNs and LTL — and Doubly So Versus Finite Automata

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arXiv:2510.19315 • ICLR 2026 Outstanding Paper Award

Fixed-precision transformers can describe the same languages as linear temporal logic and recurrent neural networks — but exponentially more compactly. Compared with finite automata, the gap is doubly exponential. The price of this compression: verifying even simple transformer properties is EXPSPACE-complete, placing fundamental limits on mechanistic interpretability.

AT A GLANCE

Problem: Prior work shows transformers and classical formalisms can express the same languages, but says nothing about *how compactly*.

Why it matters: Succinctness governs sample efficiency, parameter efficiency, and the computational cost of verification and analysis.

Method: Formal proofs constructing witness language families and providing tight translation bounds between formalism sizes.

Result: Transformers exponentially more succinct than LTL/RNNs; doubly exponentially more succinct than DFAs; verification EXPSPACE-complete.

Evidence: Matching upper and lower bounds — all results are tight.

The Big Picture

Two formalisms that recognize the same languages are not equally useful: one may require a million states to describe what the other captures in ten rules. This notion — *succinctness* — is classical in automata and logic theory, yet was essentially unexplored for transformers.

Prior work on transformer expressivity focused on *what* can be expressed: transformers can simulate circuits, recognize star-free languages, approximate Turing machines. But knowing that an equivalence holds tells you nothing about the cost of translation. This paper fills the gap by providing the first tight succinctness hierarchy for fixed-precision transformers relative to the two most natural

competitors: linear temporal logic (LTL) and recurrent neural networks.

Why Existing Approaches Fall Short

Equivalence proofs (transformer $\equiv$ circuit / LTL / RNN) are the bread and butter of prior expressivity work. They show that the formalisms are inter-translatable, but existing translation procedures blow up doubly exponentially. This was known to be a loose upper bound, but no lower bound was known, leaving open whether a polynomial translation might exist.

This paper closes that gap. The exponential blow-up in translating a transformer to LTL or RNN is not an artifact of proof technique: it is provably unavoidable.

Core Mechanism

The witness construction builds a family of languages $\{L_n\}_{n \geq 1}$:

- **Transformer description:** Size $\text{poly}(n)$ — a constant number of attention heads using positional encodings suffices.
- **LTL formula / RNN:** Any equivalent description requires size $2^{\Omega(n)}$.
- **DFA:** Any equivalent automaton requires size $2^{2^{\Omega(n)}}$.

The key technical insight is that transformers exploit *global self-attention* to encode complex ordering constraints in $O(1)$ heads that LTL must unroll through time and automata must track via exponentially many states. Positional encodings act as a compact coordinate system unavailable to LTL and automata.

Technical Formulation

Main succinctness theorem (informal): There exists a language family $\{L_n\}$ such that:

$$\text{size}(T_n^*) = \text{poly}(n), \quad \text{size}(\phi_n^*) = 2^{\Omega(n)}, \quad \text{size}(A_n^*) = 2^{2^{\Omega(n)}},$$

where T_n^* , ϕ_n^* , A_n^* are the smallest transformer, LTL formula, and DFA for L_n , respectively.

Upper bound (translation theorem): Any fixed-precision transformer with L layers, H heads, and dimension d translates to an LTL formula of size at most $\exp(O(L \cdot H \cdot d))$. This is a single exponential — improving the prior doubly exponential bound.

Verification complexity: The problem

$\text{Model-Check}(M, \varphi) := \text{“Does transformer } M \text{ satisfy property } \varphi\text{?”}$

(for φ an LTL formula) is EXPSPACE-complete. The lower bound follows from the succinctness witness: small transformers can describe exponentially complex behaviours that require EXPSPACE to examine.

Learning or Inference Procedure

This is a purely theoretical paper. The proof proceeds via:

1. **Lower bounds:** Circuit-complexity arguments (communication complexity lower bounds on multi-party protocols) show that any LTL formula or RNN for L_n must have exponential size.
2. **Upper bounds:** Explicit transformer constructions achieving $\text{poly}(n)$ size, and explicit LTL translations achieving single-exponential blow-up from any given transformer.
3. **Complexity:** A reduction from EXPSPACE-complete problems to model-checking of small fixed-precision transformers.

What the Guarantee Says

All bounds are tight:

- The exponential succinctness advantage of transformers over LTL/RNNs is matched by the single-exponential upper bound on translation.
- EXPSPACE-completeness is tight: membership in EXPSPACE follows from the translation theorem; hardness from the reduction.

Experimental Findings

No experiments. All results are formally proved. The paper establishes a complete succinctness hierarchy:

$$\text{Transformer} \prec_{\text{exp}} \text{LTL} \equiv \text{RNN} \prec_{\text{exp}} \text{DFA},$$

where $\prec_{\text{exp}}$ denotes “exponentially more succinct than.”

Ablations and Interpretation

Why succinctness matters for ML: Succinctness directly controls the inductive bias of a hypothesis class. Transformers’ exponential compression advantage explains their remarkable parameter efficiency: they can represent complex structured dependencies in far fewer parameters than classical models.

Implications for interpretability: EXPSPACE-completeness means that mechanistic interpretability — formally verifying that a transformer satisfies a property like “never attends to personal data after a system prompt reset” — is computationally intractable in the worst case. This is not a limitation of current tools; it is a fundamental barrier.

Future directions: The authors conjecture that approximate verification (checking properties of most inputs, not all) may be tractable, and leave open whether practical heuristics can bypass the worst case.

Reference

Pascal Bergsträßer, Ryan Cotterell, Anthony W. Lin. **Transformers are Inherently Succinct.** arXiv:2510.19315, October 2025. <https://arxiv.org/abs/2510.19315>